Candidate Seat No:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Second Semester of B. Pharm. (OLD) Examination April 2016

MA- 141 Advanced Mathematics

Date: 10/05/2016(Tuseday) Time: 10.00 a.m. - 1.00 p.m. **Maximum Marks: 80**

INSTRUCTIONS:

(a) **All questions are Compulsory.**
(b) **All questions carry equal marks.**
(c) **Figures to right indicate marks.**

SECTION-I

Q.1 Attempt any TWO of the following: [20]

(A)(i) Using Leibnitz's theorem, find n^{th} derivative of function $y = x^2 3^{5x}$ [05]

(ii) Find 5^{th} derivatives of function $y = e^{2x}\cos(7x - 9)$. [03]

(iii) Find 4^{th} derivative of e^{5x}. [02]

(B)(i) Solve the differential equation $\frac{d^2y}{dt^2} + y = 0$ with initial conditions $y(0) = 1$ and $y'(0) = 0$ using Laplace transform. [05]

(ii) State first shifting property for Laplace transform and hence find Laplace transform of function $e^{-6t}\sinh 2t$. [05]

(C)(i) Verify Cauchy's mean value theorem for $f(x) = x^3$ and $2 - x$, $x \in [0, 2]$. [05]

(ii) Find Taylor's series expansion of $f(x) = x^3 - 2x + 4$ around $a = 2$. [05]

Q.2 Attempt any TWO of the following: [20]

(A)(i) Verify Lagrange's mean value theorem for $f(x) = x^3 + 5$, $x \in [-1, 3]$. [05]

(ii) Find Laplace transform of $\sin 2t - \cos 5t - t$. [03]

(iii) Find 3rd derivative of 5^{2x}. [02]

(B)(i) Find 5^{th} derivative of function $y = \frac{1}{(x-1)(x-2)}$. [05]

(ii) Find Maclaurin's series of $f(x) = \sin x$. [05]

(C)(i) Find Laplace transform of function $\cos 4t - 5t^2 + e^{9t} - 1$. [05]

(ii) Find 7^{th} derivative of function $y = \log(5x - 4)$. [03]

(iii) Does Taylor's series expansion of function $f(x)$ is also a Maclaurin's series expansion of function $f(x)$? Justify your answer. [02]

SECTION-II

Q.3 Attempt any TWO of the following: [20]

(A)(i) Solve the differential equation $2xy\,dx + (x^2 + y^2 + \sin y)dy = 0$. [05]

(ii) Solve the differential equation $\frac{d^2y}{dx^2} - 4y = e^{-2x}$ using method of variation of parameters. [05]

(B)(i) Solve the differential equation $(D^2 - 14D + 49)y = 0$ [05]

(ii) Solve the linear differential equation $\frac{dy}{dx} + 4\frac{y}{x} = x$. [05]

(C)(i) The rate at which bacteria multiply is proportional to the instantaneous number present. If the original number doubles in 3 hours, in how many hours will it 4 times? [07]

(ii) What is the order and degree of the differential equation $(\frac{d^7y}{dx^7}) - y^{\frac{1}{11}} = 0$? [03]

Q.4 Attempt any TWO of the following: [20]

(A)(i) If the temperature of a body drops from 100° to 80° in one minute when the temperature of the surrounding is 30°, what will be the temperature of the body at the end of second minute? [07]

(ii) Write the general form of exact differential equation. Also write the formula for solving it. [03]

(B)(i) Solve the differential equation $\cos x \sin y\,dx + (\sin x \cos y + y^2)\,dy = 0$. [05]

(ii) Write general form of second order homogeneous linear differential equation of constant coefficients with its all possible types of solutions. [05]

(C)(i) Solve the differential equation $(D^2 + 2D + 3)y = 0$. [05]

(ii) Solve the differential equation $\frac{d^2y}{dx^2} = x^2 + 1$. [05]

*****